AXI4 Full Learning VIP

Verification Plan, Test Plan, Traceability, and Contribution Boundary

Functional AXI4 memory-mapped profile for the self-contained VIP-to-VIP project derived from legiahuy15/AXI4-VIP. Reference: Arm AMBA AXI and ACE Protocol Specification, IHI 0022H.c.

Profile	Value
Target	Core functional correctness for a learning UVM VIP, not commercial certification
Default interface	32-bit address, 32-bit data, 4-bit ID
Optional attributes	AxCACHE/AxPROT/AxQOS/AxREGION transported, stable, compared, covered; semantics default/pass-through
Ownership	Existing unmarked code remains Huy Le. New or materially modified SystemVerilog blocks are marked //Hoang Ho.
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Validation status: the earlier response-backpressure repair was run successfully for seeds 22883092 and 68865829. The larger functional patch in this package has completed static review, but must be compiled and regressed on the QuestaSim server before publication.

1. Scope and Functional Profile

- Five independent AXI4 channels: AW, W, B, AR, and R.
- FIXED, INCR, and WRAP bursts, including INCR lengths up to 256 beats.
- Narrow and unaligned transfers with byte-lane and WSTRB checking.
- Multiple outstanding transactions, same-ID ordering, and different-ID out-of-order completion.
- AW/W independence, request/response backpressure, reset recovery, response codes, and simplified exclusive access.
- Optional AxCACHE, AxPROT, AxQOS, and AxREGION fields remain pass-through/default rather than modeling a full SoC fabric.

Sign-off gate for the learning profile

Gate	Required result	Reason
Compile/optimize	0 errors; warnings reviewed	No syntax/elaboration ambiguity
make spec_smoke	100% pass	Deterministic Arm corner vectors
Positive regression	100% pass; no unexpected UVM_ERROR/UVM_FATAL	Functional test closure
Pending state	0 unmatched/pending transactions	No silently dropped traffic
Functional coverage	>= 90% core covergroups; every core miss reviewed	Verification intent closure
Assertions/directives	Enabled positive properties exercised; no unexpected failure	Protocol checker closure

A pass result means the declared learning profile is internally consistent. It does not represent Arm or commercial VIP certification.

2. Core Functional Corrections

Area	Huy Le baseline risk	Hoang Ho correction	Directed evidence
Burst/lane helpers	Address and lane arithmetic duplicated across components	Shared Arm-style FIXED/INCR/WRAP, legal lane, and exact 4KB helpers	axi4_helper_unit_test; axi4_spec_corner_test
Unaligned first beat	Modulo lane selection could wrap into lane 0 in the same beat	First transfer uses Arm lower/upper byte-lane equations	ADDR=1, SIZE=4B -> mask 1110
4KB rule	start_offset + total_bytes approximation rejected legal page-edge WRAP	Container-based FIXED/INCR/WRAP classification	WRAP 0x0FFC -> 0x0FFC,0x0FF0,0x0FF4,0x0FF8
W channel	Old beat could repeat when WREADY remained HIGH	Advance payload immediately after each handshake	Continuous-WREADY phase in spec_corner
R channel receive	Receiver assumed one RID until RLAST	Dispatch every accepted R beat by RID	ID-aware master and both monitors
Same-ID order	Concurrent preparation could allow overtaking	Per-ID FIFO/sequence order at AR acceptance	Two same-ID reads in spec_corner
B/R dependencies	Coarse counters and same-cycle ambiguity	Transaction-aware eligible write and pre-existing AR tracking	SVA dependency checks
WSTRB/error commit	Common lane arithmetic and permissive commit	Legal-mask enforcement; failed writes do not update memory	strobe/error tests
Regression infrastructure	Old UCDB could remain and shell status could hide failure	spec_smoke, strict UVM status, clean merge/report	Makefile targets

3. Requirement Traceability

ID	Arm area	Requirement	Checker/model	Positive evidence	Status
VP-HS-01	A3.2	VALID/READY transfer and stall stability	Channel drivers/monitors + SVA	sanity and backpressure tests	Implemented/improved
VP-WR-01	A3.3	AW/W independence and B after AW + final W	wr_order + eligible-write SVA	wr_order, W-before-AW, outstanding	Corrected
VP-RD-01	A3.3/A5	R after AR; RID dispatch and ordering	Per-ID driver/monitor/SVA queues	outstanding, OOO, spec_corner	Corrected
VP-BURST-01	A3.4.1	FIXED/INCR/WRAP address and length rules	Shared helpers + constraints + SVA	burst sweep + helper/spec corner	Corrected
VP-4KB-01	A3.4.1	Burst does not cross 4KB	Exact container helper + SVA	4KB test + helper unit	Corrected
VP-LANE-01	A3.4.4	Narrow/unaligned byte lanes and WSTRB	Lane helper + slave/scoreboard/SVA	narrow, unaligned, strobe, spec_corner	Corrected
VP-LAST-01	A3.2/A3.4	Exact WLAST/RLAST and no early termination	Beat counters + monitors + SVA	burst sweep/data integrity	Implemented/improved
VP-ORD-01	A5/A6	Same-ID order; different-ID OOO permitted	Per-ID queues + optional reorder	outstanding, OOO, spec_corner	Implemented/improved
VP-RESP-01	A3.4.5/A7	Response codes and EXOKAY legality	Slave/scoreboard/SVA	error and exclusive tests	Implemented/improved
VP-BP-01	A3.2	Request and response backpressure	Delay knobs + stability SVA	backpressure tests + fixed seeds	Implemented/improved
VP-ATTR-01	A4/A8	Optional attributes transported/stable	Transaction/interface/coverage	cache_prot test	Default/pass-through

4. Test Plan

Group	Tests	Primary intent
Basic/random	sanity, random	Five-channel flow, transaction matching, read-back
Burst/lane	all_burst_type, burst_sweep, narrow, unaligned, strobe	Address progression, beat count, LAST, legal lane subsets
Hoang Ho gate	helper_unit, spec_corner, 4kb_boundary	Exact helper vectors, continuous WREADY, page-edge WRAP, same-ID order
ID/order	outstanding, out_of_order, ooo_demo, back_to_back	Multiple outstanding, same-ID order, different-ID OOO
Timing	backpressure, response_backpressure, wr_order_demo, wlast_before_aw	All five stalls and AW/W legal timing combinations
Response/exclusive	error_response, exclusive, exclusive_fail, illegal_exclusive, exclusive_demo	OKAY/EXOKAY/SLVERR/DECERR and reservation policy
Integrity/reset	data_integrity, addr_integrity, reset_mid_burst	Byte memory, address mapping, reset cleanup
Attributes	cache_prot	Pass-through/stability/coverage of optional attributes
Negative SVA lane	sva_unit	Intentional reserved burst/size/LAST/dependency/stability violations

Focused validation commands

```
cd sim
make clean
make spec_smoke
make -j8 regress NUM_RUNS=5
make cov_report
```

5. Declared Limits and Contribution Boundary

Item	Treatment	Reason / sign-off boundary
USER signals	Out of scope	Not required for the learning memory-mapped profile
ACE, AXI5, AXI-Stream, coherency	Out of scope	Different protocol families/features
AxCACHE/AxPROT/AxQOS/AxREGION semantics	Default/pass-through	No cache hierarchy, permission map, QoS arbiter, or region router
Beat-level R interleaving generation	Disabled by default	Receivers are RID-aware; default subordinate keeps waveforms deterministic
Mixed-endian conversion	Out of scope	Little-endian byte placement only
Concurrent same-address R/W visibility	Simplified policy	Conflicting tests are serialized; no universal system memory-ordering claim
Commercial compliance certification	Not claimed	Functional learning VIP with static traceability and project regression

Ownership and contribution rule

Existing Huy Le class names, package names, interface signals, public APIs, author headers, and normal Make targets are retained. Entirely new SystemVerilog files begin with **//Hoang Ho - New file**. New or materially modified blocks inside existing SystemVerilog files are identified by **//Hoang Ho**. Documentation and Makefile changes use equivalent contributor notes.

The contribution should be reviewed as an incremental functional correction, not as a rewrite of the original project. Before publishing, attach compile logs, spec_smoke results, full regression results, coverage summary, and focused waveforms for continuous WREADY, page-edge WRAP, same-ID read order, and BREADY/RREADY stalls.